

AULYN -- PHYS301: Kinematics Formula Reference

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1. 1D Constant Acceleration Formulas:

- $v = v_0 + a \cdot t$
- $x = x_0 + v_0 \cdot t + 0.5 \cdot a \cdot t^2$
- $v^2 = v_0^2 + 2 \cdot a \cdot (x - x_0)$

2. Projectile Motion:

- Horizontal velocity v_x is constant; vertical acceleration $a_y = -g$ (-9.8 m/s^2).